

1. What is the fail predicate in Prolog? How does it interact with backtracking – explain with an example. [marks: 5]

2. What will be the output for the *query* ?- *do*(*X*, [*a*, *b*, *c*])? Explain step-by-step how Prolog arrives at the answer, including all unification attempts and backtracking. [marks: 6]

do(*X*, [*X*]).

do(*X*, [_ | *T*]) :- *do*(*X*, *T*).

3. Complete the definition of the rule *sibling*(*X*, *Y*) in the following code segment; and then using the facts below, write Prolog rules for *uncle*(*X*, *Y*) and *cousin*(*X*, *Y*). Show the expected output for the queries: *uncle*(*tom*, *alice*) and *cousin*(*charlie*, *alice*). [marks: 9]

parent(*john*, *mary*).

parent(*john*, *tom*).

parent(*mary*, *alice*).

parent(*mary*, *bob*).

parent(*tom*, *charlie*).

male(*john*).

male(*tom*).

male(*bob*).

male(*charlie*).

female(*mary*).

female(*alice*).

sibling(*X*, *Y*) :- ...